Credit Card Approval Prediction

(by Using Machine Learning)

Project Description:

Credit Card Approval Prediction Using Machine Learning is an AI-powered web application that automates the credit card approval process by predicting whether an applicant is eligible for a credit card based on their financial and personal information. The system uses Machine Learning algorithms such as Logistic Regression, Decision Tree, Random Forest, and XGBoost to analyse historical applicant data and generate accurate approval predictions. The best-performing model is deployed using Flask, enabling users to receive real-time approval or rejection results through a simple and user-friendly interface. By reducing manual effort and improving decision-making speed, the system helps financial institutions process applications more efficiently and consistently.

Problem Statement

Manual credit card approval processes are inconsistent, slow, and prone to subjective bias, which can result in creditworthy applicants being rejected or high-risk applicants being approved. There is a need for a data-driven, automated system that can accurately classify credit card applications as approved or rejected based on applicant attributes, thereby improving decision speed, consistency, and reliability for financial institutions.

Scenarios:

Scenario 1: Standard Credit Card Application

A customer applies for a credit card by entering their employment details, annual income, education level, housing status, and family information through the web application. The system preprocesses the input data and uses the trained Machine Learning model to predict whether the application should be **Approved** or **Rejected**, displaying the result instantly.

Scenario 2: High-Risk Applicant Evaluation

A bank officer evaluates an applicant with an unstable employment history and relatively low annual income. After submitting the applicant's details, the system analyses multiple financial and demographic factors using the trained model and predicts a **Rejected** status, helping the bank minimize potential financial risk and make informed lending decisions.

Scenario 3: Bulk Customer Screening

A financial institution processes numerous credit card applications daily. Bank employees enter each applicant's information into the system, which rapidly evaluates every application using the deployed Machine Learning model. The application provides immediate approval

predictions, significantly reducing manual verification time, improving operational efficiency, and maintaining consistent decision-making across all applications.

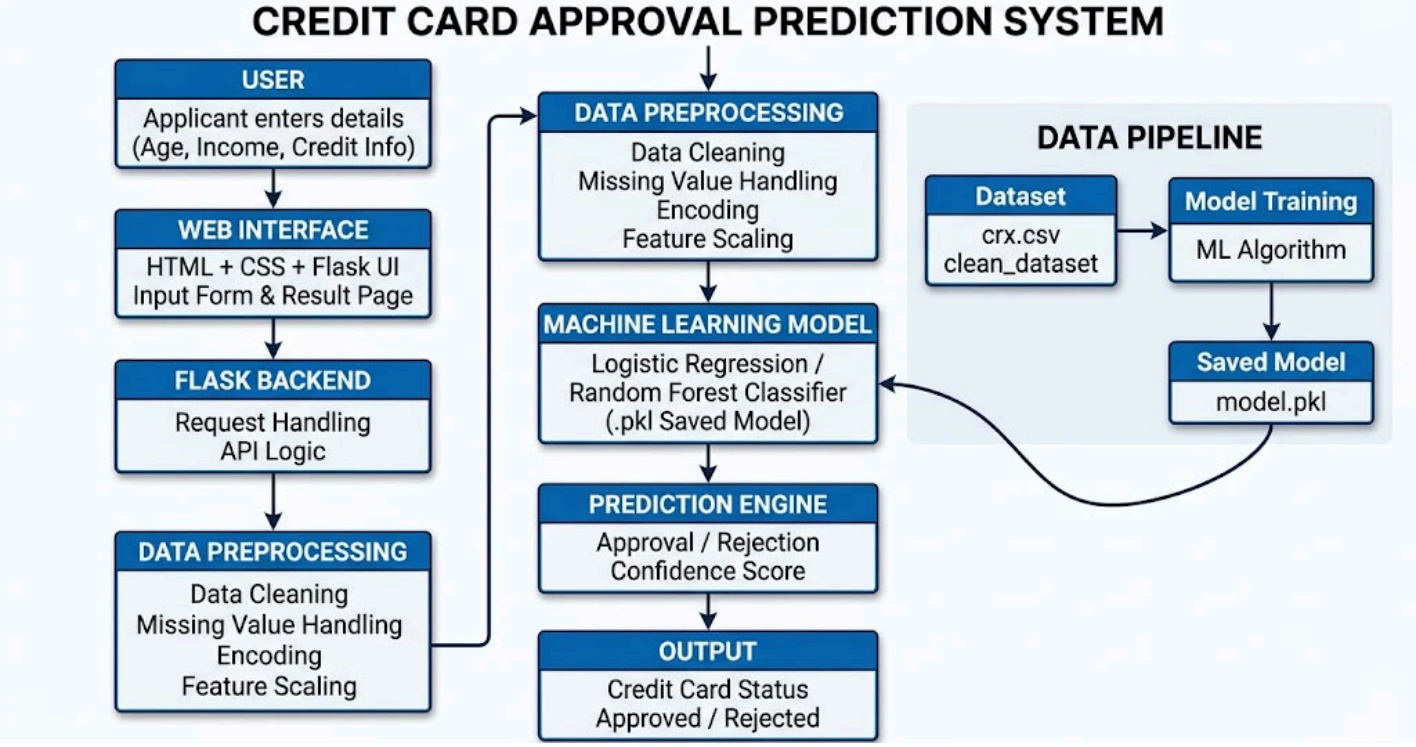
Objectives

- Automate the credit card approval prediction process using Machine Learning.
- Improve decision-making speed and reduce manual effort.
- Predict whether a credit card application is approved or rejected with good accuracy.
- Provide real-time predictions through a Flask-based web application.
- Enhance user experience with an easy-to-use and responsive interface.

Scope

The scope of this project is to develop a machine learning-based system that predicts whether a credit card application will be approved or rejected based on applicant details. The system helps financial institutions automate the initial screening process, reduce manual effort, and support faster decision-making. The project includes data preprocessing, model training, prediction, and a user-friendly web interface for entering applicant details and viewing results. Future enhancements may include real-time banking integration and advanced fraud detection features.

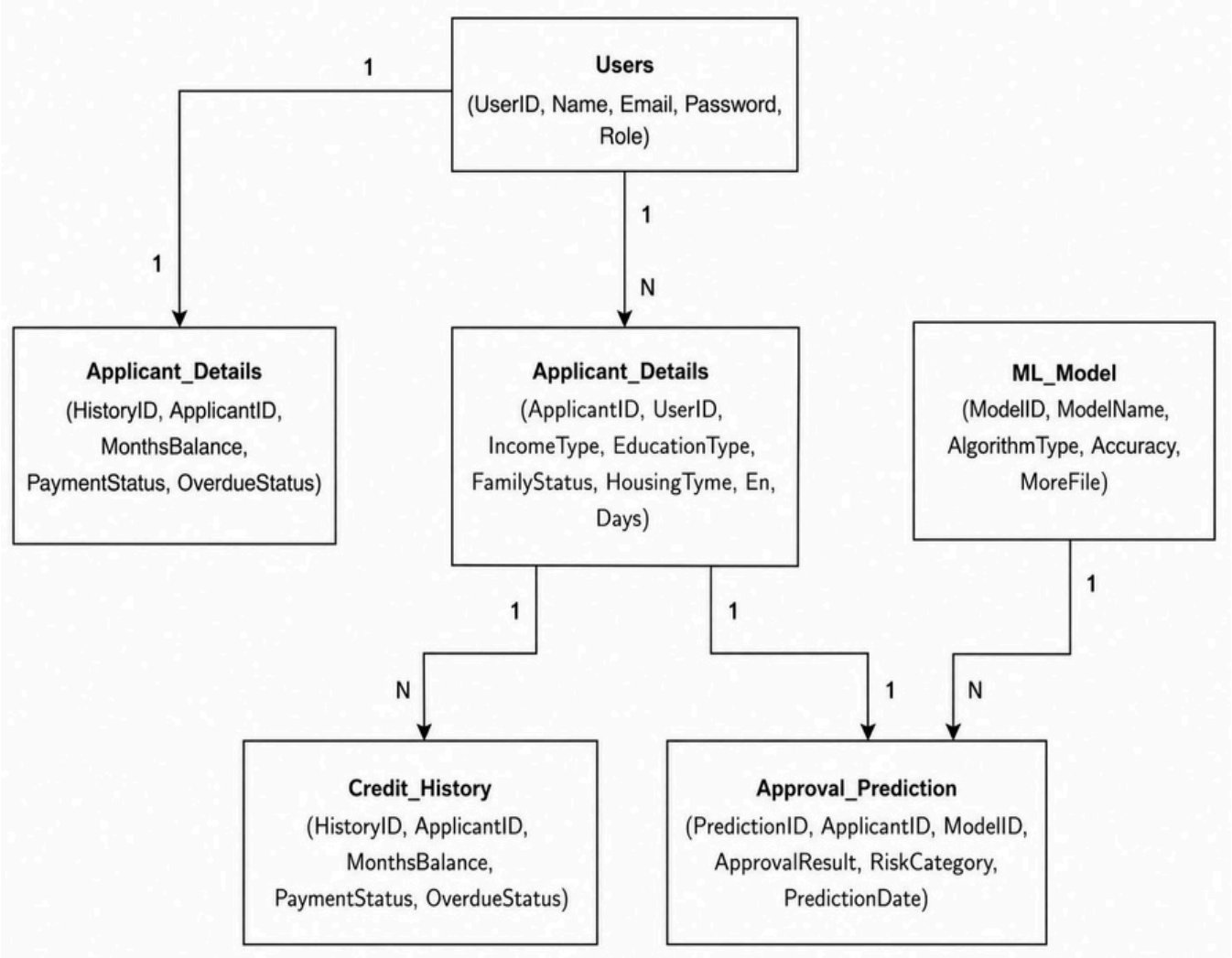
Technical Architecture:
Architecture Flow



ER Diagram

Description:

The Entity Relationship (ER) diagram of the Credit Card Approval Prediction System represents the relationship between applicants, credit card applications, and machine learning prediction results. The Applicant entity stores the personal and financial details of customers, such as age, gender, income, occupation, and credit history. Each applicant can submit one or more Credit Card Applications, which contain application-related information and credit details. The submitted application data is processed by the machine learning model, and the generated output is stored in the ML Prediction entity, which includes the prediction result (Approved/Rejected), confidence score, model details, and prediction date. The relationship between these entities helps organize applicant information, application records, and prediction outcomes efficiently, providing a structured view of the credit card approval decision-making process.



Pre-requisites

- Python Programming Proficiency (Python 3.x)
- Machine Learning Fundamentals — scikit-learn, XGBoost
- Data Handling Libraries — Pandas, NumPy
- Data Visualization — Matplotlib / Seaborn (for EDA)
- Web Framework Knowledge — Flask, for deployment of the prediction interface
- Version Control with Git

Project Workflow:

Milestone 1: Data Collection and Understanding :

- Activity 1.1: Dataset Collection:Collected credit card application dataset containing applicant and financial details.
- Activity 1.2: Dataset Loading:Loaded the dataset into Python using Pandas for analysis.
- Activity 1.3: Data Exploration:Analyzed dataset structure, features, and basic statistics.
- Activity 1.4: Data Quality Check:Identified missing values, duplicates, and inconsistencies.
- Activity 1.5: Feature Understanding:Studied important attributes affecting credit card approval prediction.

Milestone 2: Data Preprocessing and Feature Engineering

- Activity 2.1: Data Cleaning

  Handled missing values, duplicates, and inconsistent data.
- Activity 2.2: Data Transformation

  Converted categorical features into numerical format.
- Activity 2.3: Feature Engineering

  Selected and optimized important features for prediction.
- Activity 2.4: Dataset Preparation

  Split the processed data into training and testing sets for model development.

Milestone 3: Model Building

- Activity 3.1: Model Selection
  Selected suitable machine learning classification algorithms for credit card approval prediction.
- Activity 3.2: Model Training
  Trained the model using the preprocessed dataset.
- Activity 3.3: Model Evaluation
  Evaluated model performance using accuracy and other evaluation metrics.
- Activity 3.4: Model Saving
  Saved the trained model as a .pkl file for prediction and deployment.

Milestone 4: Model Evaluation and Selection

- Activity 4.1: Evaluate each trained model on the test set using Accuracy, Precision, Recall, and F1-Score.
- Activity 4.2: Generate a Confusion Matrix for each model to analyse true/false positive and negative predictions.
- Activity 4.3: Compute the ROC-AUC score to assess the models' ability to discriminate between classes.
- Activity 4.4: Compare all models and select the best-performing algorithm for deployment. Based on the evaluation results (detailed in Section 5 — Results), the XGBoost Classifier achieved the highest accuracy of 89.86% and was selected as the final model for deployment.

Milestone 5: Deployment

- Activity 5.1: Save the best-performing trained model (XGBoost) using joblib for reuse in the web application.
- Activity 5.2: Build a Flask-based web application with an HTML/CSS front-end that collects applicant details through an application form.
- Activity 5.3: Integrate the trained model with the Flask backend (app.py) to generate real time approval predictions via a /predict API endpoint.
- Activity 5.4: Deploy the application on Render, making it publicly accessible as a live web service.
- Activity 5.5: Test the live application end-to-end to verify correct functioning, from form submission to prediction display.

The application has been successfully deployed and is live at the following URL:

hhttps://credit-card-approval-prediction-1-n63c.onrender.com/

Full implementation details, source code, and screenshots of the deployed web application are provided in Section 6 (Web Application).

Methodology

Dataset Description

Attribute	Details
Dataset Name	Credit Card Approval Dataset (crx.csv / clean_dataset.csv)
Type	Structed
Target Variable	Credit Card Approval Status(accepted/rejected)
Feature Types	Categorical & Numerical attributes of attributes

Data Preprocessing Steps

- Data Loading and Inspection:

The credit card approval dataset was loaded using Pandas and analyzed using functions such as head(), info(), and describe() to understand the dataset structure, features, and statistical information.

- Missing Value Verification:

The dataset was checked for missing values and appropriate handling techniques were applied to ensure data quality.

- Duplicate Checking:

Duplicate records were identified and removed to improve data consistency and prevent biased predictions.

- Categorical Encoding:

Categorical features such as Gender, Education, Marital Status, and Credit History were converted into numerical format using encoding techniques.

- Feature Selection:

Important features influencing credit card approval decisions were selected for model training.

- Train-Test Split:

The dataset was divided into training and testing sets using train_test_split for model evaluation.

- Feature Scaling:

Numerical features were scaled using techniques such as StandardScaler to improve machine learning model performance.

Algorithms Used

- Logistic Regression (Final deployed model)
- Random Forest Classifier (Model comparison)
- Decision Tree Classifier (Model comparison)

Evaluation Metrics

- Accuracy — Overall proportion of correctly classified applications.
- Precision — Proportion of predicted approvals that were actually correct.
- Recall — Proportion of actual approvals that were correctly identified.
- F1-Score — Harmonic mean of Precision and Recall, useful for imbalanced classes.
- Confusion Matrix — Detailed breakdown of true/false positives and negatives
- ROC-AUC — Measures the model's ability to distinguish between approved and rejected classes across thresholds

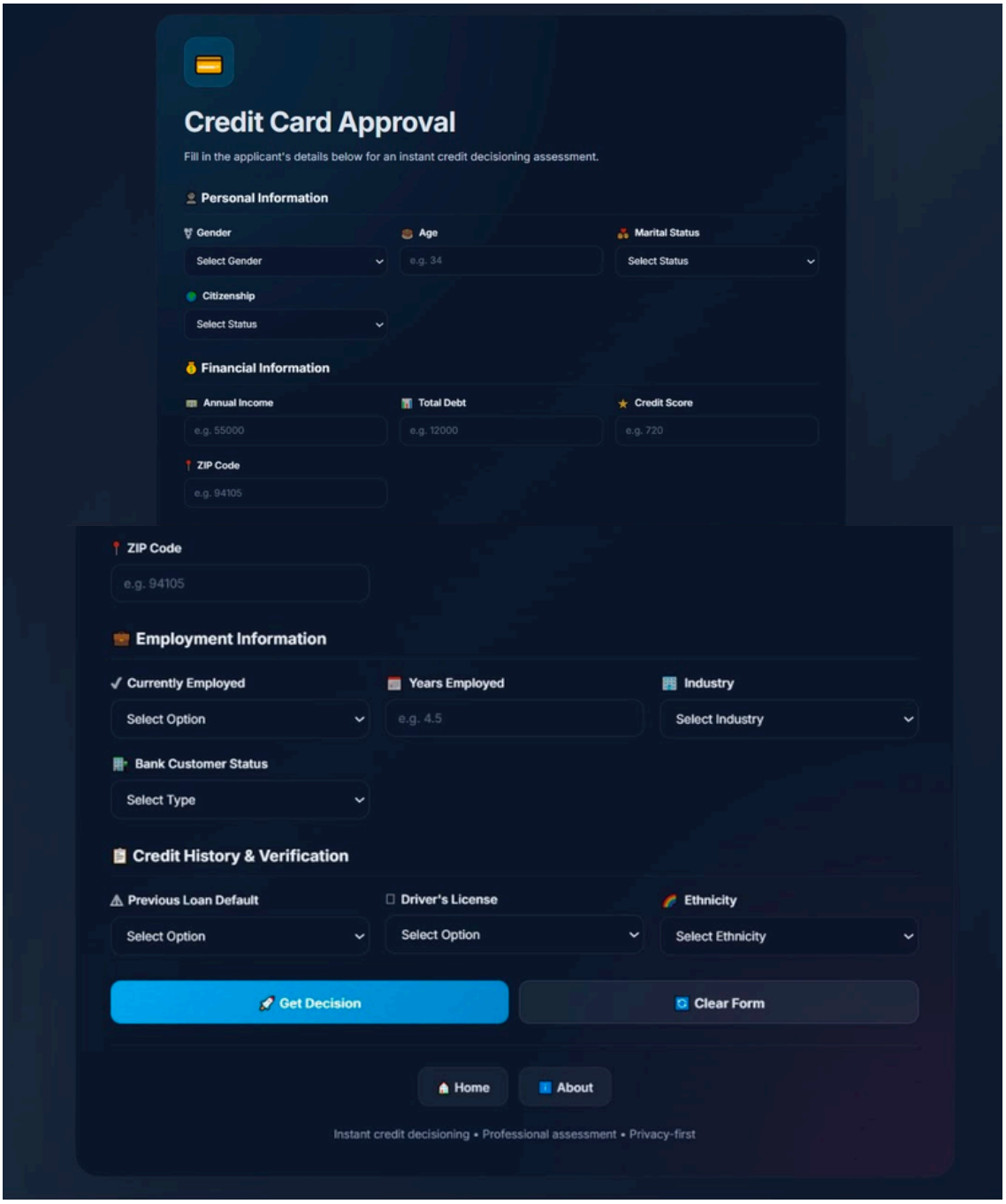
Web Application

The trained Logistic Regression model has been deployed as an interactive web application named "Credit Card Approval Prediction", built using Flask (backend) and HTML/CSS/JavaScript (frontend). The application allows users to enter applicant details through a web form and instantly predicts whether the credit card application will be Approved or Rejected, along with the prediction confidence score. The system also provides a brief explanation of the prediction based on the applicant's profile.

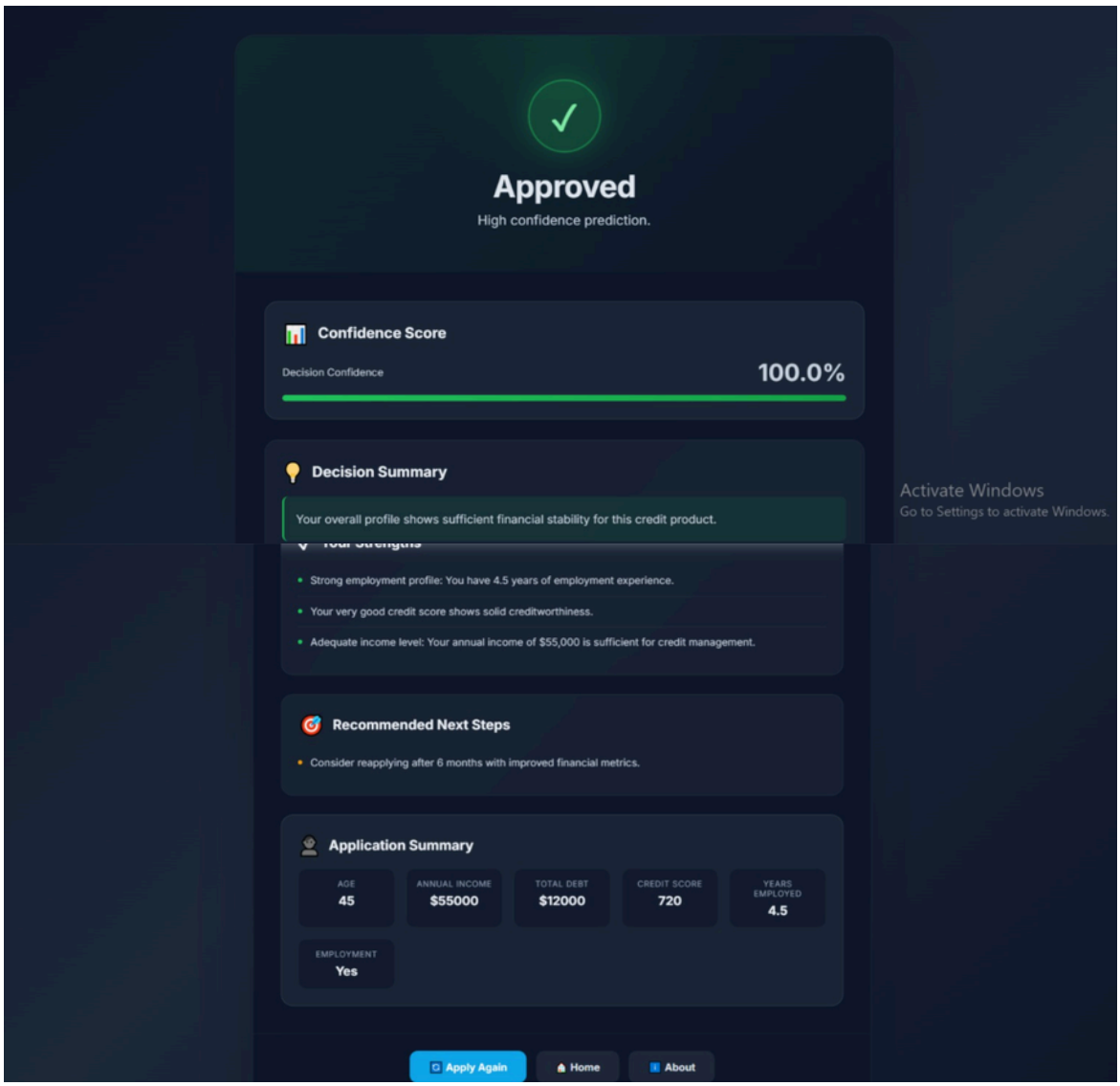
The application has been successfully deployed and is publicly accessible at:

https://credit-card-approval-prediction-1-n63c.onrender.com/

Application Form



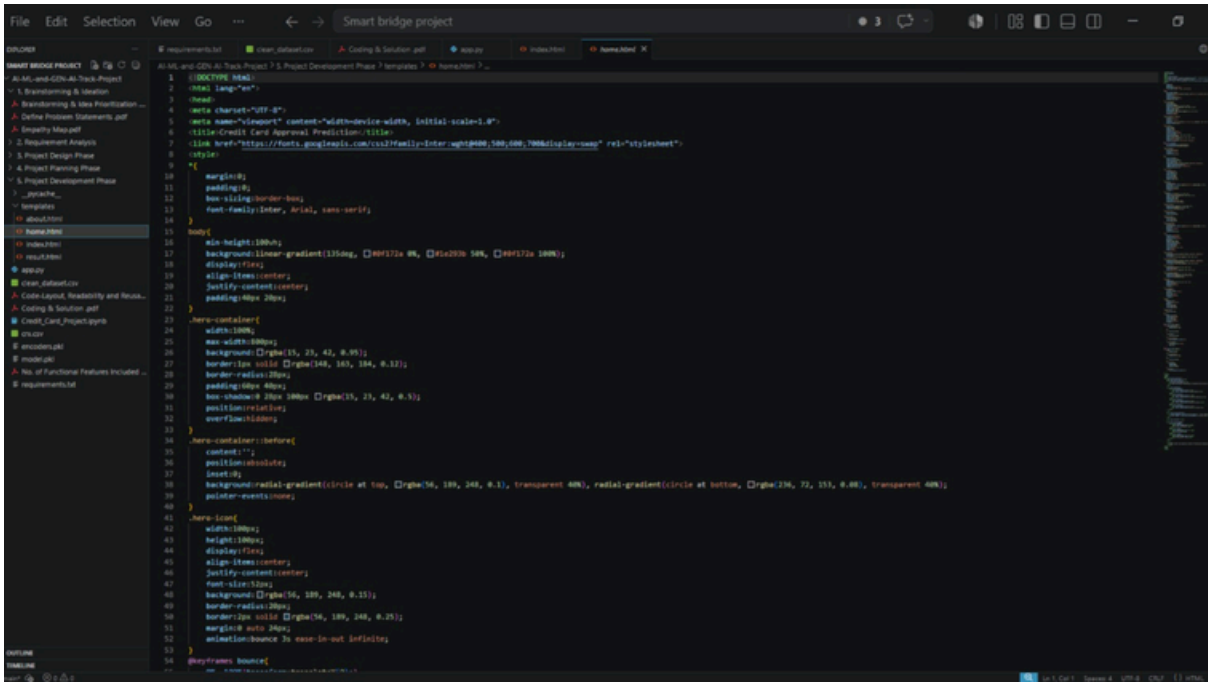
Prediction of Result

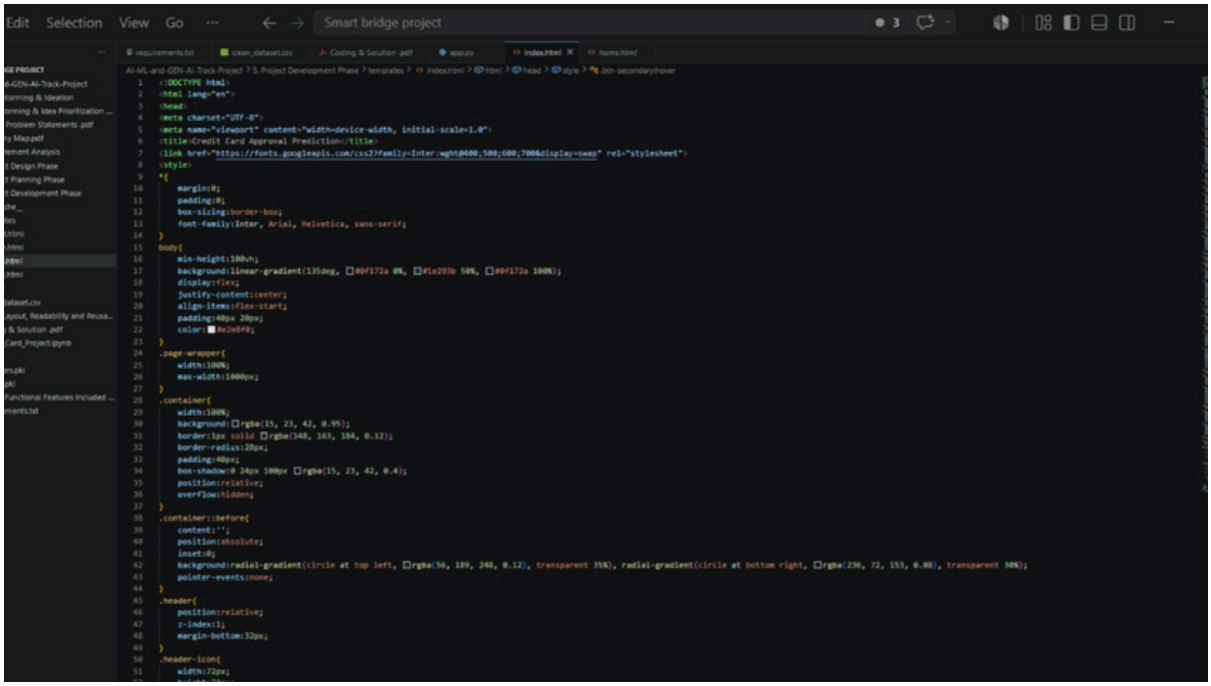


Frontend Implementation

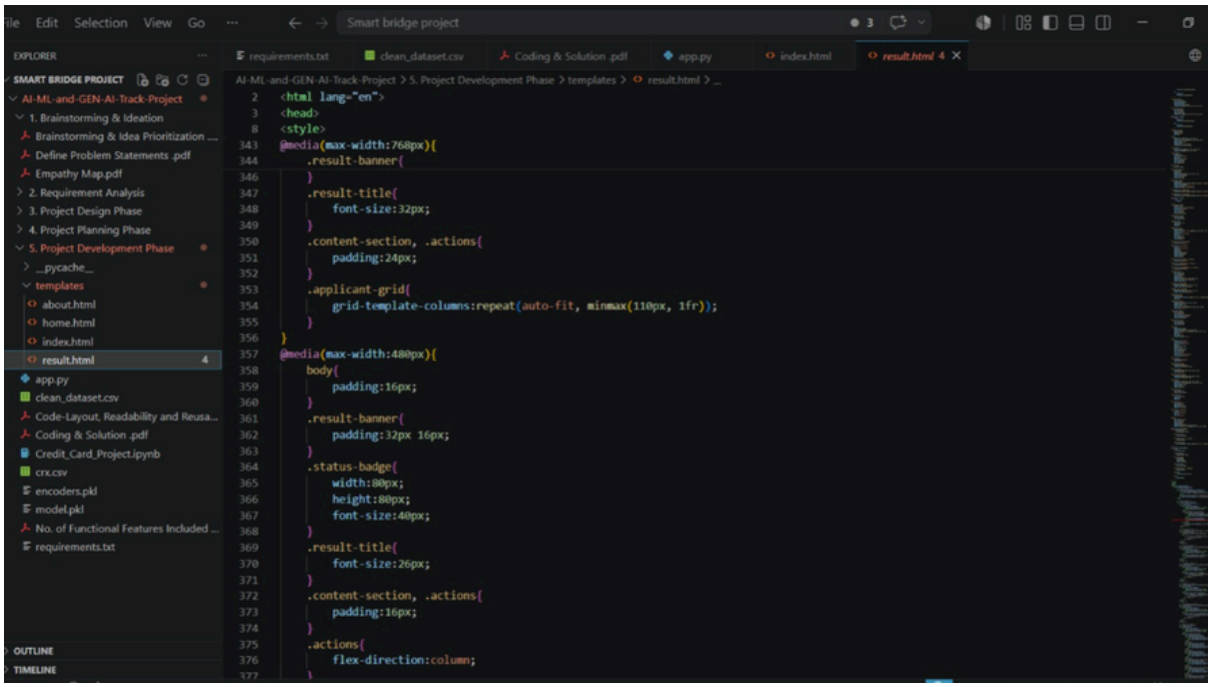
- HTML
- CSS
- JavaScript

HTML(structure)

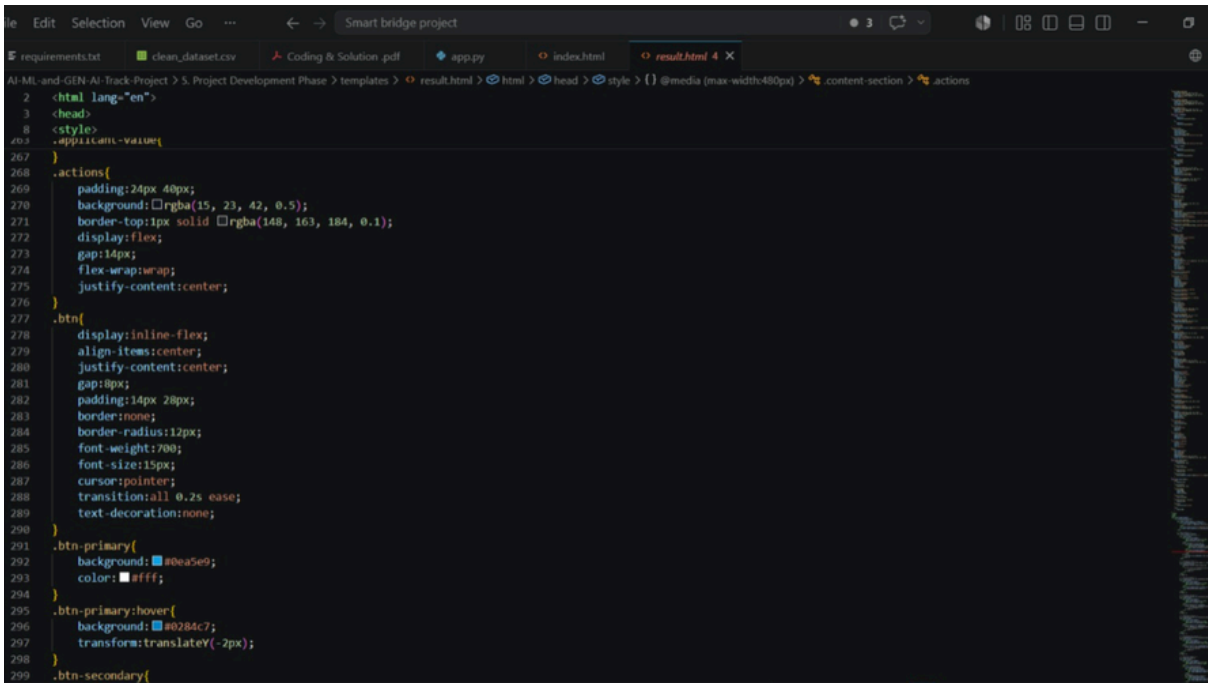




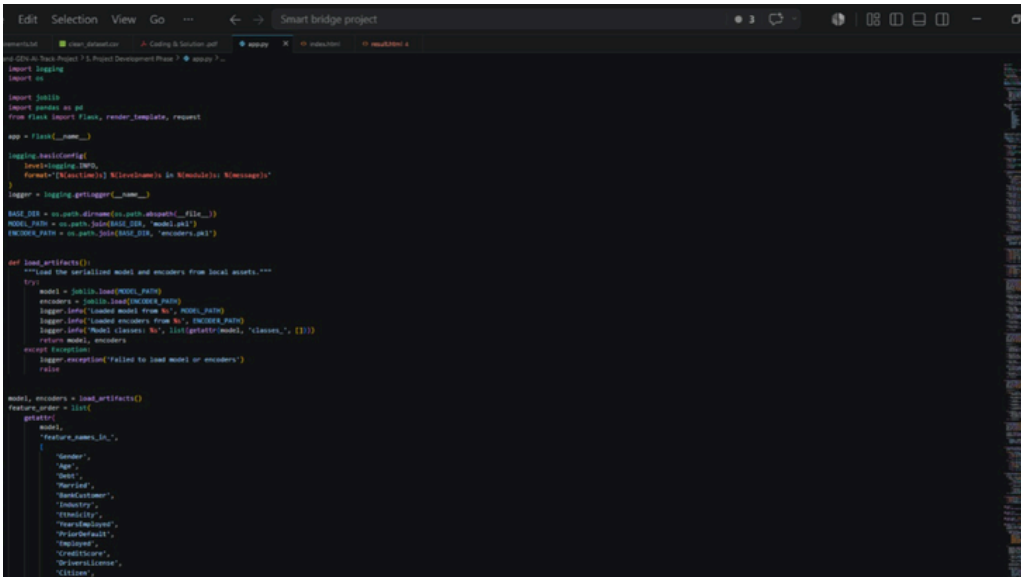
JavaScript

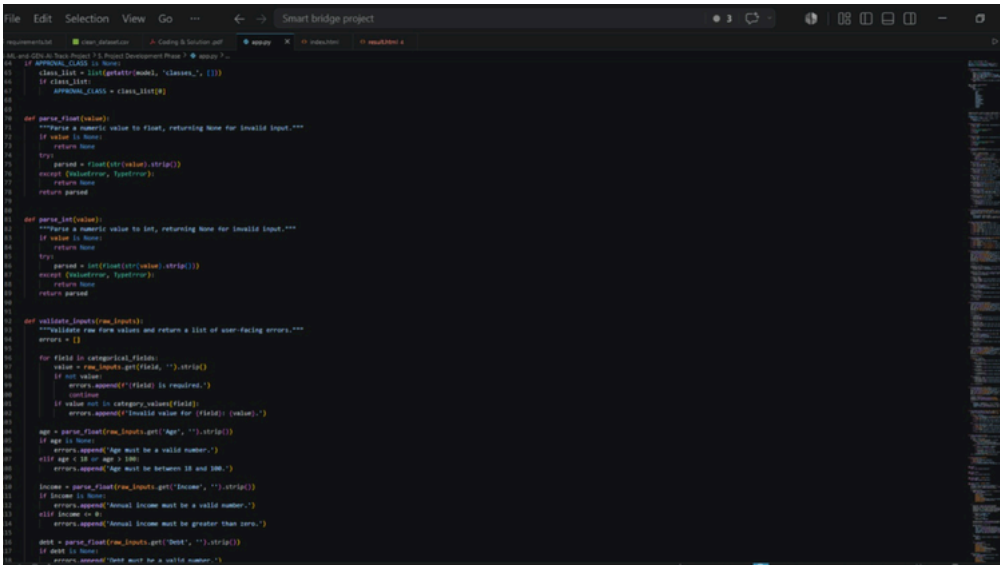


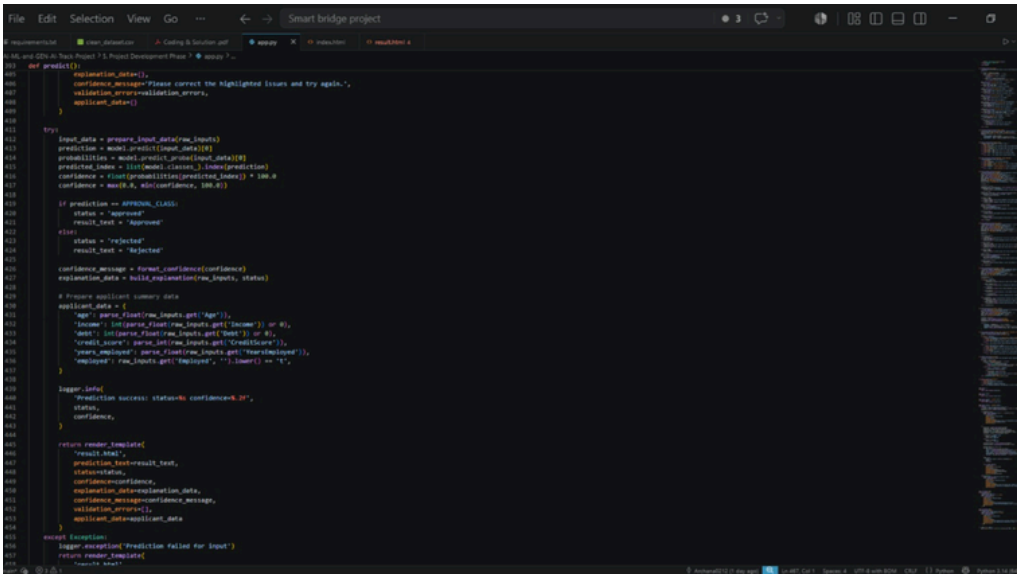
CSS



Backend Implementation Flask(app.py)







Conclusion and Future Scope

Conclusion

The Credit Card Approval Prediction using Machine Learning project successfully demonstrates how machine learning can be used to automate and improve the credit card approval process. By analyzing applicant information such as age, income, debt, employment status, credit score, marital status, and other financial attributes, the developed model accurately predicts whether a credit card application is likely to be Approved or Rejected. Multiple classification algorithms, including Logistic Regression, Decision Tree, and Random Forest, were evaluated, and Logistic Regression was selected as the final model for deployment. The model was integrated into a Flask-based web application with an interactive user interface that provides real-time predictions, confidence scores, and personalized explanations. This system helps reduce manual effort, improves consistency in decision-making, minimizes human bias, and enables faster credit approval assessments. Overall, the project demonstrates that machine learning is an effective solution for enhancing the efficiency, accuracy, and reliability of the credit card approval process.

Future Scope

The proposed Credit Card Approval Prediction system can be further enhanced with several advanced features to improve its performance and real-world usability. Future improvements include:

- Integrating the system with real-time banking and financial databases for live applicant verification
- Enhancing prediction accuracy by training on larger and more diverse datasets.
- Incorporating advanced machine learning and deep learning models for improved performance.
- Adding fraud detection mechanisms to identify suspicious or high-risk applications.
- Implementing user authentication and role-based access for secure access to the application.
- Deploying the application on cloud platforms with database support for better scalability and accessibility.
- Providing detailed decision explanations and personalized recommendations to help applicants improve their approval chances.